Visual-based Automatic Drone Detection and Position Calculation

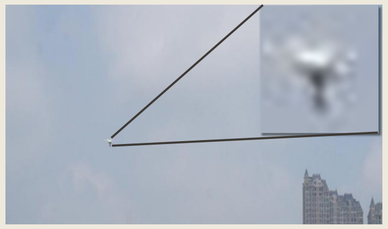
Bachelor of Engineering (Honours) In Internet-of-Things with a Secondary major in Artificial Intelligence and Data Analytics Final Year Project

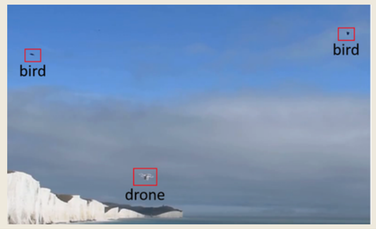
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Introduction

- **Increasing Demand:** Rapid drone proliferation demands robust detection systems.
- **Challenges:**
  - Drones at 100-1000m present only 36-100 pixels, which is challenging for traditional computer vision
  - Other flying objects like birds need to be distinguished from drones to avoid false alarm.

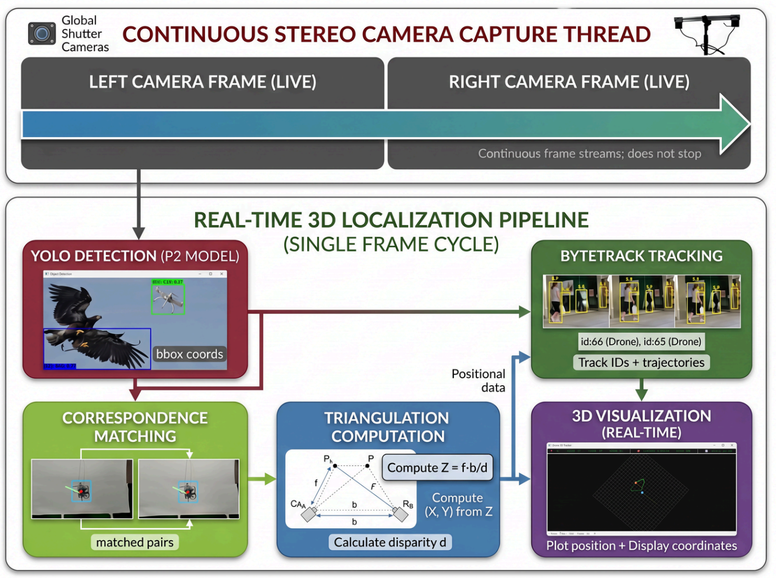




Objectives

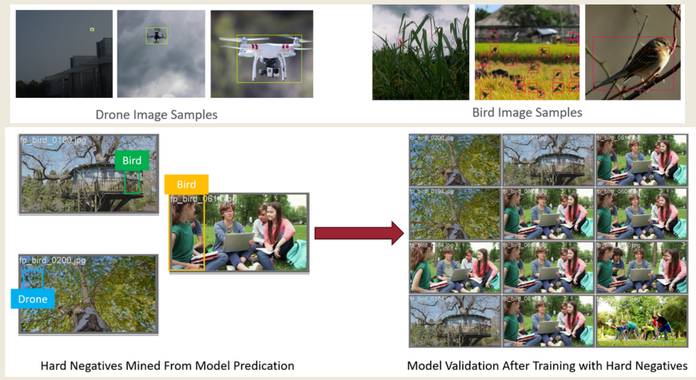
- Enhanced Detection Model (>90% precision, >80% recall)
- Multi-Object Tracking (stable trajectories, occlusion handling)
- 3D Localization (<10% error)
- Drone v.s. Bird Classification

Method	Advantages	Disadvantages
Radar	All weather, wide coverage	Fails on small drones, expensive
RF Detection	Penetrates obstacles	Requires known frequencies
Acoustic	Cost-effective, better efficiency	Limited range, noise interference
Vision ✔	Rich information, cost-effective	Weather dependent

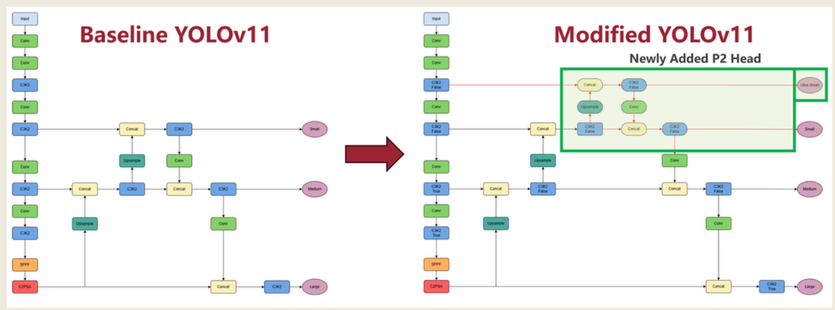


Methodology

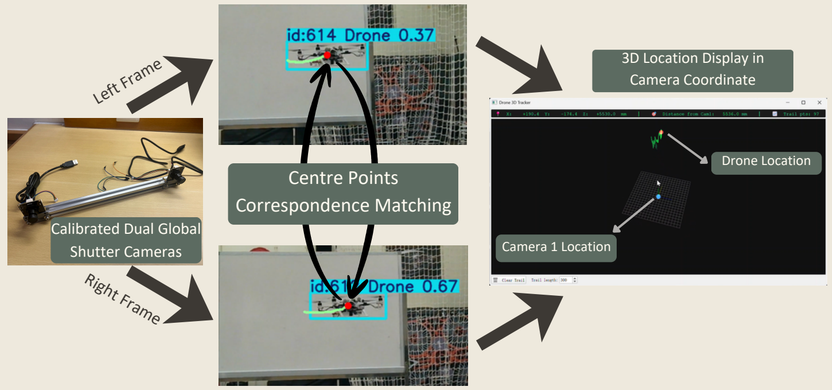
Dataset Construction



Model Architecture



3D Localization



Conclusion

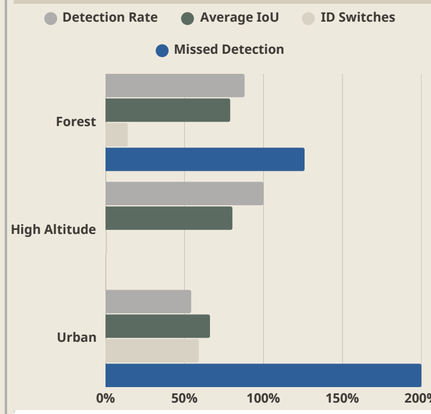
Achievements:

- P2-Head YOLOv11 architecture
- 10,397-image dual-class dataset
- End-to-end detection → tracking → 3D localization

Future Work:

- Larger model variants (YOLOv11-small/medium)
- Dataset expansion (nighttime, weather conditions)
- Extended camera network for wider coverage

Results and Discussion



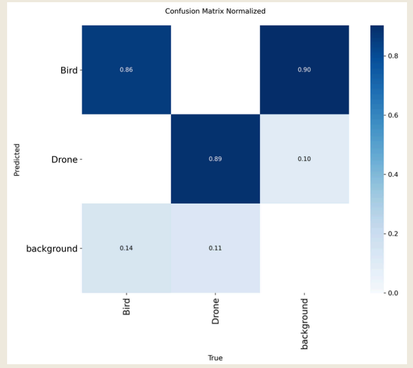


Table 2 Summary of Model Performance

Model	AP50	AP50-95	Precision	Recall	Small	Medium	Large
UAV_small	0.60	0.34	0.99	0.98	-	-	-
Bird_small	0.89	0.64	0.88	0.85	-	-	-
UAV_large	0.85	0.52	0.93	0.80	-	-	-
Dual_v2	UAV:0.85 Bird:0.89	UAV:0.50 Bird:0.55	UAV:0.92 Bird:0.89	UAV:0.79 Bird:0.81	Drone Bird 0.39	Drone Bird 0.57	Drone Bird 0.65
Dual_P2	**Drone:0.89 Bird:0.90**	**Drone:0.53 Bird:0.54**	**Drone:0.92 Bird:0.89**	**Drone:0.82 Bird:0.81**	**Drone Bird 0.48 0.39**	**Drone Bird 0.67 0.55**	**Drone Bird 0.74 0.63**

Table 3: Overall Performance Comparison

Metric	Baseline	P2 Head	Δ
mAP50-95	0.510	0.524	+2.7%
AP_small	0.413	0.440	+6.5%
AP_medium	0.622	0.613	-1.4%
AP_large	0.704	0.687	-2.4%

Table 4: Per-Class Small Object Detection

Class	Baseline	P2 Head	Δ
Bird	0.385	0.396	+2.9%
Drone	0.440	0.485	+10.2%

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